

Title: TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods

Aim

The project, which resulted in the TRIPOD+AI statement, aims to promote the complete, accurate, and transparent reporting of studies that develop or evaluate the performance of clinical prediction models.

Key objectives of the project include:

- **Harmonising Reporting:** It provides a consolidated set of recommendations for prediction model studies, regardless of whether they employ regression modelling or machine learning (AI) methods.
- **Assisting Stakeholders:** The project aims to help a wide range of users - including authors, peer reviewers, editors, policymakers, and patients - to better understand and critically appraise the data, methods, findings, and conclusions of AI-driven research.
- **Facilitating Evaluation and Implementation:** Complete reporting is intended to make it easier for others to evaluate the quality of a model and successfully implement it into clinical pathways to support decision-making.
- **Addressing Modern Methodological Standards:** The project updates the original 2015 guidance to incorporate contemporary standards for fairness, open science, and reproducibility, which have become increasingly important with the rise of machine learning in healthcare.
- **Reducing Research Waste:** By providing minimum reporting recommendations, the project seeks to eliminate "avoidable research waste" caused by incomplete or unusable reports, ensuring that research outputs are valuable to the scientific community and safe for patients.

To achieve these aims, the project developed an expanded 27-item checklist that supersedes previous versions and provides detailed explanations for each reporting recommendation.

Theoretical Framework / Key Concepts

The theoretical framework of the TRIPOD+AI statement centers on providing a harmonised reporting structure for clinical prediction models, regardless of whether they are developed using traditional regression modelling or modern machine learning (AI) methods.

The key concepts and definitions used in this framework include:

1. Core Prediction Model Concepts

- **Prediction Models:** Tools used to estimate an outcome value or risk, primarily to support clinical decision-making. They generally fall into two categories:
 - **Diagnostic:** Estimating the probability that a particular health condition is currently present.
 - **Prognostic:** Estimating the probability that a particular outcome will occur in the future.
- **Predictors (Features/Covariates):** Characteristics measured at an individual level (e.g., age, blood pressure) or group level used as inputs for the model.
- **Outcome (Target/Label):** The specific diagnostic or prognostic event the model is designed to predict.

2. Machine Learning and AI Definitions

- **Artificial Intelligence (AI):** A field of computer science focused on algorithms capable of performing tasks that typically require human intelligence.
- **Machine Learning (ML):** A subfield of AI concentrated on models that learn and make decisions from data without being explicitly programmed.
- **Hyperparameters:** Specific values that control the model development or learning process, often requiring "tuning" to find the best settings for a strategy.

3. Data and Validation Framework

- **Training (Development) Data:** The dataset used to build or "train" the prediction model.

- **Evaluation (Test) Data:** Distinct data used to estimate the model's performance. It must have no overlap in participants with the training data to ensure an unbiased assessment.
- **Internal Validation:** The process of evaluating model performance on the same population used for development, often using techniques like cross-validation or bootstrapping.

4. Measures of Model Performance

The framework evaluates a model's "predictive accuracy" through three primary pillars:

- **Discrimination:** How well the model differentiates between individuals who have the outcome and those who do not (e.g., using the C-statistic or AUROC).
- **Calibration:** The agreement between the values estimated by the model and the actual observed outcomes, best assessed via a calibration plot.
- **Clinical Utility:** An assessment of the model's practical value in a clinical setting (e.g., decision curve analysis).

5. Emerging Pillars of Reporting

TRIPOD+AI integrates several contemporary concepts into its reporting requirements:

- **Fairness:** The property that a model does not discriminate against individuals or groups based on protected attributes like age, sex, or race/ethnicity.
- **Open Science:** Principles of transparency and collaboration, including the public registration of studies and the sharing of protocols, data, and analytical code.
- **Class Imbalance:** Situations where the frequency of individuals with and without the outcome is unequal, often requiring specific methods to prevent model bias.
- **Patient and Public Involvement (PPI):** Integrating diverse stakeholder perspectives into the design and conduct of research to improve clinical acceptability and health equity.

Methodological Concepts

Methodological concepts within the TRIPOD+AI statement define the essential procedures for the development and evaluation of clinical prediction models. These concepts are primarily organised within the Methods section of the 27-item checklist (items 5–17).

Data and Participant Management

- **Data Sources and Representativeness:** Researchers must describe the sources of data (e.g., routine care registries, randomised trials) and provide a rationale for their use, focusing on the representativeness of the data for the target population.
- **Participant Selection:** Methodology involves clearly defining eligibility criteria, the study setting (e.g., primary vs. secondary care), and the location of centres. It also requires detailing how any treatments received by participants were handled during model building.
- **Data Preparation:** This includes all pre-processing and quality-checking steps, with a specific requirement to report whether these processes were consistent across different sociodemographic groups to ensure fairness.

Variable Definition and Analysis

- **Outcome and Predictors:** Both the outcome (the event being predicted) and the predictors (the inputs or features) must be clearly defined. This includes specifying the time horizon for predictions and describing the qualifications of assessors if the measurements require subjective interpretation.
- **Sample Size Justification:** A key methodological requirement is providing an explanation and justification for the study size, including any specific sample size calculations used to ensure the study is sufficiently powered.
- **Handling Missing Data:** The report must describe the methods used to handle missing data and provide reasons for omitting any participants or variables from the analysis.

Model Building and Evaluation

- **Analytical Procedures:** The methodology requires a detailed description of how data were used (e.g., partitioning for development and evaluation), how predictors were handled (e.g., rescaling or transformation), and the specific model-building steps, including hyperparameter tuning.
- **Validation Strategies:**
 - **Internal Validation:** Evaluating performance on the same population used for development via techniques such as cross-validation or bootstrapping.
 - **Model Evaluation:** Estimating predictive accuracy through three pillars: discrimination (e.g., c-statistic), calibration (e.g., calibration plots), and clinical utility (e.g., decision curve analysis).
- **Class Imbalance and Fairness:** If specific methods were used to address class imbalance (unequal outcome frequency) or to ensure model fairness across groups, these must be explicitly stated along with their rationale.
- **Model Updating:** For evaluation studies, methodology includes describing any recalibration or updating performed to improve the model's performance in a new setting or for specific subgroups.

Transparency and Reproducibility

- **Model Output and Specification:** Researchers must specify the type of output (e.g., probabilities or classifications) and provide the full prediction model (via formula, code, or API) to allow for independent third-party evaluation and implementation.
- **Open Science Practices:** Methodological transparency is supported by reporting the availability of the study protocol, registration information, and the analytical code used for data cleaning and model building.

Assumptions

The TRIPOD+AI statement does not prescribe or assume a specific methodology for how a prediction model should be developed or evaluated; instead, it focuses on the minimum reporting requirements needed to make such research transparent and usable. However, the framework is built upon several key operational and scope-related assumptions:

- **Non-Generative Scope:** The guidance is primarily aimed at non-generative models. While its principles can help drive transparency in generative AI studies (like those using large language models), the current statement was developed without explicit consideration for generative approaches.
- **Data Independence:** A fundamental assumption regarding data handling is that evaluation (test) data must be entirely distinct from the data used to train the model or tune hyperparameters. The sources specify that there should be no overlap in participants between the training and evaluation datasets to ensure an unbiased assessment of performance.
- **Representativeness:** The framework assumes that both training and evaluation data should ideally be representative of the population in which the model is intended to be used. If the data used to develop the model do not adequately represent the full diversity of the target population, the sources assume the resulting model may not perform as expected for the missing groups.
- **Legacy Consistency:** TRIPOD+AI assumes the ongoing validity of the rationale provided in the TRIPOD 2015 explanation and elaboration document. Because many reporting items have remained the same or underwent only minimal changes, that document is still considered essential for providing the background logic for current reporting standards.
- **Stakeholder Interpretation:** The project assumes that complete and accurate reporting is a prerequisite for stakeholders - including clinicians, regulators, and patients - to critically appraise the quality and safety of a prediction model before it is implemented in a clinical pathway.

Methodology

The development of the TRIPOD+AI statement followed a structured methodological approach recommended by the EQUATOR Network, involving literature reviews, expert surveys, and consensus meetings.

The methodology consisted of the following key phases:

1. Initiation and Oversight

A steering group was convened to oversee the guideline's development. The initiative was announced in 2019, registered with the EQUATOR Network, and followed a formal study protocol published in 2021. The consensus methods adhered to the ACCORD recommendations for accurate consensus reporting.

2. Candidate Item Generation

The steering group generated an initial list of 65 unique candidate items. This process involved:

- Using the original TRIPOD 2015 checklist as a baseline.
- Identifying additional items from other relevant reporting guidelines (such as CLAIM, SPIRIT-AI, and CONSORT-AI) and recent literature on machine learning.
- Incorporating findings from systematic reviews that evaluated the reporting and methods of machine learning-based prediction models.

3. Modified Delphi Exercise

A two-round, anonymous Delphi survey was conducted to rate the importance of the candidate items.

- Participants: Panellists were recruited globally to ensure disciplinary diversity, including researchers (statisticians, data scientists, machine learning experts), clinicians, journal editors, regulators, and patients.
- Round 1: 170 participants rated 65 items. Items with $\geq 70\%$ agreement for being "desirable" or "essential" were carried forward.
- Round 2: 200 participants rated a refined list of 59 candidate items.

4. Patient and Public Involvement (PPIE)

An online meeting was held with nine members of a patient and public involvement group. This step was used to revise the checklist for clarity and ensure that the project aims and scope were acceptable to end-users and the general public.

5. Consensus Meeting and Finalisation

The process concluded with an online consensus meeting attended by 28 multi-stakeholder participants.

- The group discussed a subset of 17 items that either lacked consensus from the Delphi rounds, were newly introduced, or had undergone significant rewording.
- Participants voted to include or exclude these items using a formal polling function to form the final 27-item checklist.

Main Findings

The development of the TRIPOD+AI statement led to several key findings regarding the current state of clinical prediction model research and the necessary standards for reporting when using machine learning (AI).

1. Deficiencies in Current Reporting and Methodology

Systematic reviews conducted during the project revealed significant issues in the existing literature:

- Poor Conduct and Methodology: Many studies are plagued by deficiencies in study design, data collection, and poor methodology.
- Incomplete Reporting: Key details are frequently missing, leaving studies at a high risk of bias and making them difficult to appraise or implement.
- Lack of Open Science: Prediction model studies rarely adhere to open science practices, such as sharing protocols, data, or analytical code.
- Susceptibility to "Spin": Research is often prone to overinterpretation or "spin," which can mask flaws and lead to safety concerns if implemented in clinical pathways.

2. The Need for an AI-Specific Update

The authors found that the previous standard (TRIPOD 2015) was insufficient for modern needs:

- ML Complexity: Unlike traditional regression, machine learning approaches (like deep learning or random forests) often do not result in a simple equation, and the predictors used can remain unclear, necessitating additional reporting considerations.
- Methodological Advances: Since 2015, there have been significant advances in sample size guidance and a greater recognition of the need for fairness, reproducibility, and open science.

3. The Core Output: A Harmonised 27-Item Checklist

The project successfully established a new global standard:

- **TRIPOD+AI Checklist:** This is an expanded 27-item checklist (comprising 52 subitems) that provides minimum reporting recommendations regardless of whether regression or machine learning was used.
- **Superseding Old Guidance:** A major finding of the consensus process is that the TRIPOD+AI checklist now supersedes the TRIPOD 2015 checklist, which should no longer be used.

4. New Pillars of Reporting

The statement identifies three critical areas that were previously underreported:

- **Fairness:** There is a new, heavy emphasis on ensuring models do not discriminate against groups based on attributes like age, sex, or race. This is now embedded throughout the checklist (items 3c, 5a, 7, 8a, 8b, 9c, 12f, 14, 20b, 23a, 25, and 26).
- **Open Science:** To improve integrity, researchers are now expected to report on study registration, protocol availability, and data/code sharing.
- **Patient and Public Involvement (PPI):** The project found that involving diverse stakeholder perspectives is essential for clinical and public acceptability, leading to a specific new reporting item for PPI.

5. Benefits of Adherence

The study concludes that adherence to these standards provides several benefits:

- **Reduced Research Waste:** Complete reporting prevents "avoidable research waste" by ensuring studies are usable by others.
- **Improved Trust and Safety:** Transparent reporting creates more trust among the public and allows regulators and clinicians to verify a model's safety before clinical deployment.
- **Facilitated Implementation:** By providing full model specifications (including formulas or code), researchers enable third-party evaluation and real-world implementation.

Limitations

The sources identify limitations regarding the current state of clinical prediction research, the scope of the TRIPOD+AI statement itself, and the specific constraints researchers must address when reporting their findings.

1. Scope and Technical Limitations of TRIPOD+AI

- **Non-Generative Focus:** At the time of development, foundation and large language models (LLMs) like ChatGPT were not explicitly considered; the guidance is primarily aimed at non-generative models.
- **Need for Iteration:** Because AI and machine learning (ML) methods advance rapidly, the statement will require periodic updating to remain relevant.
- **Reporting vs. Quality Appraisal:** The checklist is strictly a tool for transparent reporting and does not prescribe how to develop a model, nor is it a quality appraisal tool for assessing risk of bias (for which tools like PROBAST are recommended).

2. Limitations in Current Research and Data

- **Poor Conduct and Reporting:** Systematic reviews have found that many prediction model studies are poorly conducted, use deficient methodology, and are incompletely reported with key details missing.
- **Masked Flaws:** Incomplete reporting can mask significant flaws in study design or conduct, which could cause real-world harm if a biased model is implemented in a clinical pathway.
- **Complexity of ML Models:** Unlike traditional regression, the complexity of methods like random forests or deep learning often means models do not result in a simple equation, and the predictors used can remain unclear.
- **Data Representation and Fairness:** Representation of minoritised or underserved groups in a dataset does not alone guarantee fairness. If training data is not representative of the target population, the model may fail to perform as expected for the missing groups.

3. Required Reporting of Study-Specific Limitations

The TRIPOD+AI checklist (Item 26) specifically requires authors to discuss the limitations of their own studies and how these factors affect bias, statistical uncertainty, and generalisability. Key areas to address include:

- Non-representative samples.
- Insufficient sample size.
- Overfitting of the model to training data.
- Missing data and how it was handled.
- Uncertainty regarding input data: Authors must discuss how poor quality or unavailable data should be assessed if the model is implemented in practice.

Implication of Practise

The TRIPOD+AI checklist serves as a rigorous framework for practitioners to ensure that clinical prediction models are safe, transparent, and operationally reliable before they are deployed in real-world forecasting contexts.

Pre-Deployment Auditing and Full Specification

The 27-item checklist acts as a pre-deployment audit tool; any gap in reporting is viewed not just as an omission, but as a potential risk to patient safety. Practitioners are advised to demand full model specifications, including reproducible code, formulas, or accessible APIs. Without these, independent verification is impossible, and the model cannot be effectively stress-tested or recalibrated for new clinical settings.

Comprehensive Performance Evaluation

Beyond standard discrimination metrics like auROC, the framework emphasizes two other pillars of performance:

- Calibration: Practitioners must require calibration plots to ensure predicted probabilities match observed outcomes. Models with poor calibration can lead to systematically biased risk estimates, requiring recalibration when moved to populations with different outcome prevalences.
- Clinical Utility: Decision curve analysis is the primary tool for determining if using a model actually improves patient outcomes compared to simpler strategies. This analysis should be run against local data to confirm a "net benefit" exists for the specific local case mix.

Fairness and Subgroup Integrity

To prevent discriminatory behaviour, performance metrics - including discrimination, calibration, and utility - must be stratified across subgroups based on age, sex, ethnicity, and other relevant attributes. If performance gaps are found, practitioners should apply subgroup-specific recalibration or restrict the model's use to validated populations. This evaluation must continue during post-deployment monitoring as populations evolve.

Operational Protocols and Open Science

The framework transitions technical hurdles into clinical decisions:

- Missing Data: Deployment plans must include a protocol for missing predictors, defining whether to delay prediction, use imputation, or escalate to manual review.
- Class Imbalance: If models used oversampling or weighting during development, practitioners must use Bayes' rule corrections to ensure probability outputs reflect true clinical risk in the deployment environment.
- Open Science: Pre-registering study protocols is encouraged to prevent "spin" or the post-hoc modification of analysis to produce more favourable results.

Strategic Design and Stakeholder Communication

The checklist should be used prospectively during the initial data collection design to ensure that necessary variables (such as those for fairness) are recorded and that sample sizes are sufficient. Furthermore, practitioners must create plain-language summaries for clinical stakeholders that explicitly state the model's limitations, validated populations, and circumstances where clinical judgement should override the tool.

The "Minimum Bar" Caution

While TRIPOD+AI ensures transparency, it is not a guarantee of methodological quality. Practitioners should use it as a minimum reporting bar in conjunction with quality-appraisal tools like PROBAST to fully assess the risk of bias before implementation.